GRADE 10 · MATHEMATICS ENGLISH MEDIUM

QUESTION PAPER

Model Paper — 05

Third Term Test (Full Syllabus)

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Time: 3 hours **Total marks:** 100

Name: ..... Class: ..... Date: .....

Instructions:

- This question paper consists of two parts, **A** and **B**. Answer **all** the questions.
- Answer the 20 questions in **Part A** in the space provided on this paper itself. (2 marks for each question)
- Answer the 6 questions in **Part B** on the separate sheets of paper provided. (10 marks for each question)
- Show the steps of your calculations clearly. Draw graphs on the graph paper provided.
- Where necessary, take the value of π as $\frac{22}{7}$. The use of calculators is not allowed.

Part	A	B1	B2	B3	B4	B5	B6	Total
Marks	/40	/10	/10	/10	/10	/10	/10	/100

Part A

Answer all the questions in the space provided on this paper. (2 × 20 = 40 marks)

1. Find the square root of 0.0196.

Answer: .....

2. Simplify: $(2\frac{1}{2} - 1\frac{1}{3}) \div 3\frac{1}{2}$

Answer: .....

3. Expand and simplify: $(x - 5)(x + 5) - (x - 2)^2$

Answer: .....

4. Factorise: $x^2 - 11x + 28$

Answer: .....

5. Factorise completely: $3a^3 - 12a$

Answer: .....

6. Simplify: $\frac{x^2 - 4x}{x^2 - 16}$

Answer: .....

7. Find the perimeter of a sector of radius 14 cm and central angle 45°.

Answer: .....

8. In triangle PQR, PQ = PR. The magnitude of $P\hat{Q}R$ is twice that of $Q\hat{P}R$. Find the magnitude of $Q\hat{P}R$.

Answer: .....

9. If $\log_2 x + \log_2 4 = 5$, find the value of x.

Answer: .....

10. If lg 3 = 0.4771, write the value of lg 0.03 in logarithm-table form (with a positive decimal part).

Answer: .....

11. Which term is the first negative term of the arithmetic progression 40, 37, 34, ...?

Answer: .....

12. If n(A) = 20, n(B) = 12 and $B \subset A$, write $n(A \cap B)$ and $n(A \cup B)$.

Answer: .....

13. Find the total interest received when a sum of Rs. 20 000 is deposited for 2 years at an annual compound interest rate of 5%.

Answer: .....

14. A piece of work is completed in 12 days by 10 workers working 8 hours a day. Find the number of days taken to complete it by 12 workers working 10 hours a day.

Answer: .....

15. A train 200 m long travels at a speed of 72 km per hour. Find, in seconds, the time taken for it to pass an electricity pole.

Answer: .....

16. In the formula $y = \frac{2x + 3}{x - 1}$, make x the subject.

Answer: .....

17. A bag contains 3 red balls, 5 white balls and 2 black balls, all of the same size. Find the probability that a ball taken at random is not white.

Answer: .....

18. If the mean of the numbers 3, 7, x, 10, 12 is 8, find the value of x.

Answer: .....

19. In parallelogram ABCD, $A\hat{D}B = 35°$ and $A\hat{B}D = 60°$. Find the magnitude of $B\hat{C}D$.

Answer: .....

20. The curved surface area of a cylinder of radius 7 cm is 440 cm^2. Find the height of the cylinder.

Answer: .....

Part B

Answer all the questions on separate sheets of paper. (10 × 6 = 60 marks)

1. (a) Simplify: $\frac{3}{x^2 - x - 6} + \frac{1}{x + 2}$ (3 marks)

(b) The price of 2 pens and 3 books is Rs. 410. The price of 3 pens and 2 books is Rs. 340. Taking the price of a pen as Rs. p and the price of a book as Rs. b, form a pair of simultaneous equations and solve them to find the price of a pen and the price of a book. (4 marks)

(c) Solve: $(x - 2)(x + 3) = 14$ (3 marks)

2. An incomplete table of values prepared for drawing the graph of the function $y = \frac{1}{2}x^2 - 2$ is given below.

x	−4	−3	−2	−1	0	1	2	3	4
y	6	.....	0	−1.5	−2	.....	0	2.5	6

(a) Complete the blank spaces in the table. (1 mark)

(b) Draw the graph taking 10 small squares to represent 1 unit along the x-axis and 10 small squares to represent 1 unit along the y-axis. (3 marks)

(c) Using the graph,

(i) write the minimum value of the function and the coordinates of the turning point. (1 mark)

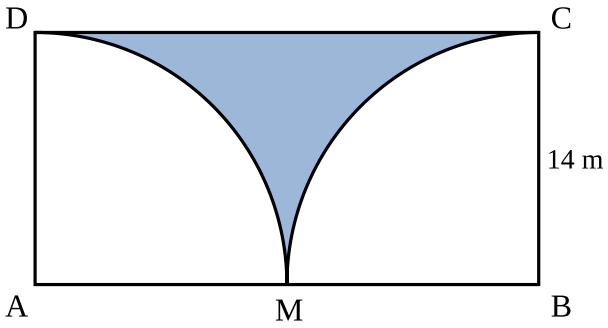
(ii) write the roots of the equation $\frac{1}{2}x^2 - 2 = 0$. (1 mark)

(iii) by drawing a suitable straight line, find approximately the roots of the equation $\frac{1}{2}x^2 - 2 = 3$. (2 marks)

(iv) write the range of values of x for which $y \leq 2.5$. (1 mark)

(d) The graph is translated upwards by 2 units along the y-axis. Write the function of the resulting graph. (1 mark)

3. In the rectangular plot of land ABCD shown in the figure, AB = 28 m and BC = 14 m. M is the midpoint of AB. The arcs DM and CM are circular arcs of radius 14 m drawn with A as the centre and B as the centre respectively.



(a) Find the area of the sector ADM. (2 marks)

(b) Find the area of the shaded region. (2 marks)

(c) Find the perimeter of the shaded region. (3 marks)

(d) Find the total cost of planting grass on the two sectors at Rs. 200 per m^2 and paving the shaded region with stones at Rs. 1 000 per m^2. (3 marks)

4. In parallelogram ABCD, the bisector of $D\hat{A}B$ meets the side DC at E, and the bisector of $A\hat{B}C$ meets the side DC at F.

(a) Prove that the triangle ADE is an isosceles triangle. (3 marks)

(b) If AD = 7 cm and AB = 12 cm, find the length of EC. (2 marks)

(c) Prove that AE and BF intersect perpendicularly. (3 marks)

(d) Using the data in (b), find the length of EF. (2 marks)

5. The marked price of a television set is Rs. 120 000.

(a) A discount of 12% is given when it is bought for cash. Find the amount to be paid when it is bought for cash. (2 marks)

(b) If it is bought on the instalment plan, 25% of the marked price must be paid as the initial payment, and 12 monthly instalments of Rs. 8 500 each must be paid. Find, as a percentage of the marked price, the amount paid in excess of the marked price under the instalment plan. (3 marks)

(c) If the trader had bought the television set for Rs. 88 000, find the profit percentage he makes when he sells it for cash. (2 marks)

(d) Find the price that should be marked so that a profit of 25% on the cost price is still obtained after giving a discount of 12%. (3 marks)

6. A water tank can be filled in 6 hours by pipe A alone and in 4 hours by pipe B alone. The full tank can be emptied in 12 hours by pipe C.

(a) Find the fraction of the tank filled in one hour when both pipes A and B are opened. (2 marks)

(b) Find, in hours and minutes, the time taken to fill the empty tank using both pipes A and B. (2 marks)

(c) If all three pipes A, B and C are opened at the same time, find the time taken to fill the empty tank. (3 marks)

(d) Pipes A and B were opened for one hour into the empty tank, and then B was closed and C was opened. Find the further time taken for the tank to be completely filled. (3 marks)

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